SECURITY ENGINEERING RESEARCH POSTER

# Reproducible Regression Harness for Machine-Learning Security Tools

## Contract-Validated Adapters Catching Cross-Repo Breakage Before It Ships

github.com/poojakira/mlsec-benchmark-suite (public, default branch main, primary language Python). Apache-2.0 at this historical commit (later revisions may use other terms) • Python 3.12 • HEAD 5c20f1da9602eb162efe8c63158da52b11d61f60 • verified 2026-10-01

EVIDENCE POSTURE

**Verified, scoped, reproducible**

Claims are constrained to the repository evidence snapshot and stated limitations.

| 1.00 | 3/0 | 83 across 10 modules | 82 passed, 1 skipped |
|---|---|---|---|
| HF-SCANNER F1 | TP / FP | Test functions | Historical pinned CI |

SYSTEM VIEW

## Architecture and Threat-Control Flow

**Trust Boundary**

**Input surface**
Data, models, requests, policies, or artifacts entering the tested path.

**Control surface**
Repository-specific validation, analysis, and fail-closed checks.

**Evidence surface**
Findings, metrics, tests, logs, or reproducible artifacts.

This diagram describes the tested repository path, not a universal deployment guarantee.




## Research Contribution

**A contract-validated regression harness that wraps ML security tools in typed in-process adapters, runs them against versioned fixtures, and fails pytest on schema drift - with a hard, labeled separation between REAL product adapters and a synthetic smoke-test scaffold.**

## Verified Evidence

| | | | |
|---|---|---|---|
| EV-01 | Statement coverage: 90.07% | EV-02 | Real fixtures: 5 |
| EV-03 | Real adapters: gated CI | EV-04 | Known-bad flagged: 100 |
| EV-05 | Known-good clean: 100 | EV-06 | Smoke scaffold (synthetic): 50 |
| EV-07 | VERIFIED_AT_SNAPSHOT - Real HF-scanner adapter: precision=recall=F1=1.0 on 5 fixtures - README verified real run; tests/test_hf_scanner_adapter.py end-to-end (no mock); total_tp=3,total_fp=0,total_fn=0. | EV-08 | VERIFIED_AT_SNAPSHOT - 83 test functions across 10 test modules; current Python 3.12 CI recorded 82 passed, 1 skipped with 90.07% statement coverage - repository README and test tree at HEAD 5c20f1da9602eb162efe8c63158da52b11d61f60. |
| EV-09 | VERIFIED_AT_SNAPSHOT - JSON-schema contracts + versioned fixtures; pytest fails on drift - README how-it-works; schemas/ + contracts/ present. | EV-10 | UNSUPPORTED (disclaimed by repo) - run-smoke per-category numbers - README: seeded-PRNG synthetic scaffold for plumbing, NOT measurements. Shown as synthetic only. |

## Method and Control Path

01 Load (fixtures)
02 2 Adapt (in-proc)
03 3 Run (tool)
04 4 Validate (schema)
05 5 Assert (contract)
06 6·7 Report (Markdown)

## Limitations and Residual Risk

- Validates contracts/format, not real-world accuracy.
- run-smoke numbers are synthetic (seeded PRNG).
- Fixture coverage is finite; gaps possible.
- Adapters can go stale if a tool changes internals.
- Real adapter gating requires sibling install.

## Reproducibility

```
Evidence: results/real_hf_scanner.json, METHODOLOGY.md,
tests/ Measured code snapshot:
5c20f1da9602eb162efe8c63158da52b11d61f60
```

## Evidence Provenance

**Claim discipline**
Metrics come from the repository's verified metrics and research brief.

**Scope discipline**
Benchmarks are reported only at their documented fixture, dataset, or test scope.

**Reproduction**
PDF and PNG are generated locally from committed evidence without paid services.

**Evidence-backed security engineering**

README, verified metrics, tests, CI, and committed artifacts remain authoritative. Poster layout emphasizes architecture, controls, evidence, limitations, and reproduction rather than unsupported efficacy claims.

36 x 48 in
PDF + PNG parity